

SeokHyeon (Lucas) Kong

CONTACT INFORMATION	Moore 102, University of Pennsylvania 200 South 33rd Street, Philadelphia, PA 19104	Personal Website: lucasshkong.github.io Email: shkong@seas.upenn.edu
RESEARCH INTERESTS	My research interests lie in <i>Quantum Computer Architecture</i> , currently focusing on Quantum Error Correction.	
EDUCATION	University of Pennsylvania , Philadelphia, PA	Aug 2026 – Present
	• Ph.D. in Computer and Information Science • Advisor: Prof. Gushu Li	
	University of Pennsylvania , Philadelphia, PA	Aug 2026 – Present
	• M.S. in Computer and Information Science	
	Sungkyunkwan University (SKKU) , Suwon, Korea	Mar 2020 – Feb 2026
PUBLICATIONS	• B.S. in Electronic and Electrical Engineering • B.S. in Advanced Semiconductor Engineering – System Architecture Track • <i>Graduated with Summa Cum Laude</i> (GPA: 4.36/4.5)	
	The Pennsylvania State University , University Park, PA	Fall 2024
	• Exchange Student in Computer Engineering • <i>Dean's List</i> (Grade: 4.0/4.0)	
	[1] Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy SeokHyeon Kong, Donghwan Kim, Euseong Seo, and Kiwan Maeng CAL 2025 – IEEE Computer Architecture Letters	
	[2] Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system SeokHyeon Kong and Hayoung Oh JKIICE 2023 – Journal of the Korean Institute of Information and Communication Engineering	
POSTER PRESENTATION	[3] Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices SeokHyeon Kong, Chanhee Park, Jang-il Moon, and Denny Dahl Qcenter – Quantum Information Research Support Center, Korea	
EXPERIENCE	University of Pennsylvania , Philadelphia, PA	Aug 2026 – Present
	<i>Graduate Research Fellow in the Penn Quantum System Lab</i>	
	Hanwha Systems , Seongnam, Korea	Jan 2026 – Jul 2026
	<i>Software Engineer in the Avionics System R&D Team</i> • Collaborated with The Boeing Company on ELAD (Eagle Large Area Display) for F-15 aircraft.	

	IonQ , Remote <i>Research Mentee with Dr. Denny Dahl</i>	Aug 2025 – Sep 2025
	- Addressed the Instant Insanity problem by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods [3]. - Solved the Four Corners Map Coloring problem using a Quantum Approximate Optimization Algorithm (QAOA) with two different encoding schemes [3].	
	The Pennsylvania State University , University Park, PA <i>Undergraduate Researcher with Prof. Kiwan Maeng</i>	Aug 2024 – Sep 2025
	- Conducted GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differential privacy [1]. - Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality [1].	
TEACHING EXPERIENCE	Academic Tutor , SKKU <i>Intro to Electromagnetism</i>	Fall 2023
	Led weekly Q&A sessions and problem-solving discussions.	
	Teaching Assistant , SKKU <i>Engineering Mathematics 2</i>	Summer 2023
	Prepared and organized class materials, including exams.	
HONORS AND AWARDS	Graduate Fellowship and Research Stipend Award , UPenn Summa Cum Laude , SKKU Director's Award – Quantum Hackathon, QCenter President's List , SKKU Dean's List , SKKU Korea–U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000), KIAT Student Success Scholarship , SKKU	2026 2026 2025 2024, 2025 SP2023, FA2023, SP2024, FA2024, SP2025 FA2024 FA2023
SKILLS	Software : Lean, Qiskit, CUDA, C, Python, MATLAB, LaTeX Hardware : System Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute	
LEADERSHIP AND SERVICE	Executive Board , President's List (Honor Society at SKKU) Qiskit Advocate , IBM Quantum Squad Leader , Republic of Korea Army (Military Service)	2025 2025 Apr 2021 – Oct 2022
REFERENCES	Gushu Li, Ph.D. Assistant Professor of Computer and Information Science University of Pennsylvania Kiwan Maeng, Ph.D. Assistant Professor of Computer Science and Engineering The Pennsylvania State University Denny Dahl, Ph.D. Senior Director of Field Engineering IonQ	Email: gushuli@seas.upenn.edu Email: kvm6242@psu.edu Email: denny.dahl@ionq.co